

Arjuna JEE 2025

Mathematics

DPP: 5

Permutations and Combinations

- Q1** The number of ways that 8 beads of different colours be string as a necklace is
 (A) 2520 (B) 2880
 (C) 5040 (D) 4320
- Q2** The number of ways in which 7 girls can stand in a circle so that they do not have same neighbours in any two arrangements?
 (A) 720 (B) 380
 (C) 360 (D) None of these
- Q3** How many words are formed if the letters of the word GARDEN are arranged with the vowels in alphabetical order?
 (A) 120 (B) 240
 (C) 360 (D) 480
- Q4** The number of arrangements of the letters of the word 'Delhi' if *e* always comes before *i* is
 (A) 40 (B) 30
 (C) 60 (D) 20
- Q5** Number of ways in which 5 men and 3 women can be seated at a round table if no two women sit together
 (A) 1240 (B) 1440
 (C) 1300 (D) 1360
- Q6** The number of ways in which 5 ladies and 7 gentlemen can be seated in a round table so that no two ladies sit together, is
 (A) $\frac{7}{2}(720)^2$
 (B) $7(360)^2$
 (C) $7(720)^2$
 (D) 720
- Q7** Number of ways in which 5 men and 3 women can be seated at a round table if two particular women must sit together, while the third one must not sit beside those two
 (A) 2880 (B) 540
 (C) 960 (D) 4990
- Q8** If all permutations of the letters of the word AGAIN are arranged as in a dictionary, then 49th word is
 (A) NAAGI (B) NAGAI
 (C) NAAIG (D) None of these
- Q9** 20 persons were invited for a party. The number of ways they and the host be seated at a circular table if two particular persons sit on either side of the host, is
 (A) 18!
 (B) $18! \times 2!$
 (C) $18!/2!$
 (D) 19!
- Q10** There are 20 persons among whom are two brothers. The number of ways in which we can arrange them around a circle so that there is exactly one person between the two brothers, is
 (A) 18!
 (B) $17! \times 2!$
 (C) $18! \times 2!$
 (D) 20!
- Q11**



There are 5 gentlemen and 4 ladies to dine at a round table. The number of ways in which they can seat themselves so that no two ladies are together, is

- (A) 2880 (B) 576
(C) 1440 (D) 480

Q12 In a class there are 10 boys and 8 girls. The teacher wants to select either a boy or a girl to represent the class in a function. The number of ways the teacher can make this selection.

- (A) 18
(B) 80
(C) $^{10}P_8$
(D) $^{10}C_8$

Q13 The number of diagonals in a polygon of n sides, is

- (A) $\frac{n(n-1)}{2}$
(B) $\frac{(n-1)(n-2)}{2}$
(C) $\frac{n(n-3)}{2}$
(D) none of these

Q14 In a Mathematics paper there are three sections containing 4, 5 and 6 questions respectively. From each section 3 questions are to be answered. In how many ways can the selection of questions be made?

- (A) 34 (B) 800
(C) 1600 (D) 9600

Q15 Out of 7 men and 4 ladies a committee of 5 is to be formed. In how many ways can this be done so as to include at least 3 ladies

- (A) 90 (B) 120
(C) 91 (D) 360

Q16 A polygon has 90 diagonals. Number of its sides is

- (A) 25 (B) 17

- (C) 15 (D) 14

Q17 The maximum number of points of intersection of 8 circles, is

- (A) 16 (B) 24
(C) 28 (D) 56

Q18 Let T_n denote the number of triangles which can be formed using the vertices of a regular polygon of n sides. If $T_{n+1} - T_n = 21$, then n equals

- (A) 5 (B) 7
(C) 6 (D) 4

Q19 The number of diagonals that can be drawn by joining the vertices of an octagon, is

- (A) 20 (B) 28
(C) 48 (D) None of these

Q20 A committee of 7 has to be formed from 9 boys and 4 girls. In how many ways can this be done when the committee consists of

- (i) exactly 3 girls?
(ii) at least 3 girls?
(iii) at most 3 girls



Answer Key

Q1 (A)

Q2 (C)

Q3 (C)

Q4 (C)

Q5 (B)

Q6 (A)

Q7 (C)

Q8 (A)

Q9 (B)

Q10 (C)

Q11 (A)

Q12 (A)

Q13 (C)

Q14 (B)

Q15 (C)

Q16 (C)

Q17 (D)

Q18 (B)

Q19 (A)

Q20 (i) 504

(ii) 588

(iii) 1632

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